AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

COMPUTER GRAPHICS

SECTION: H

Fall 2024-2025

Project Title: Spaceship System

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Date of Submission: 19th January 2025

Introduction

- **What we implemented in “Spaceship System” project**

The “Spaceship System” project implements two detailed static 2D scenes, focusing on **space** and a **futuristic city**. These scenes form the core of the project, representing two distinct yet interconnected aspects of a visionary future.

Space Environment

The space environment is designed to capture the vastness, mystery, and beauty of outer space. It features:

• **Celestial Objects**: Stars, planets, and distant galaxies scattered across a dark, infinite backdrop. These elements create a sense of scale and wonder, emphasizing the uncharted possibilities of space exploration.

• **Dynamic Effects**: Glowing stars and subtle gradients to mimic light scattering and enhance realism.

• **Spaceship Integration**: The spaceship appears as a central object, symbolizing humanity’s progress in exploring the cosmos.

The space scene serves as a metaphor for exploration, innovation, and the pursuit of knowledge beyond Earth’s boundaries. It is crafted to give users a sense of awe and curiosity about the limitless possibilities that lie beyond our planet.

Futuristic City

The futuristic city environment represents a technologically advanced urban space that highlights human creativity and ingenuity. Key features include:

• **Towering Skyscrapers**: Modern, sleek architecture illuminated by glowing neon lights. These structures symbolize progress and innovation in urban development.

• **Technological Aesthetics**: Flying vehicles, vibrant advertisements, and interactive screens contribute to the city’s cutting-edge ambiance.

• **Dynamic Urban Atmosphere**: Simulated movement and lighting effects bring the city to life, reflecting a thriving urban lifestyle.

The futuristic city is designed as a vision of harmony between advanced technology and urban living. It portrays a society where technological advancements have shaped daily life in innovative ways.

- **Why we implement in “Spaceship System” project**

The decision to implement static 2D scenes of space and a futuristic city stems from our vision of a future where **space travel and city life coexist**. This dual implementation reflects a broader narrative about human evolution, technology, and survival.

- **Coexistence of Space Travel and Urban Life**

We envisioned a world where technological advancements have made space travel a routine part of life. In this future, cities are hubs of innovation, seamlessly connected to interstellar exploration. The futuristic city highlights the progress of humanity on Earth, while the space scene showcases our aspirations to explore new frontiers. Together, these scenes depict a harmonious coexistence of terrestrial and extraterrestrial lifestyles, offering a glimpse into a world where humanity thrives both on Earth and beyond.

- **A Post-Earth Reality**

Another compelling reason for choosing these scenes is to explore an alternative scenario—a time when life on Earth has been abandoned due to environmental or existential crises. This vision aligns with themes of sustainability and survival, encouraging reflection on the importance of preserving life on Earth while pursuing advancements in technology and exploration. The futuristic city, in this context, serves as a reminder of what humanity can achieve, while the space scene symbolizes the search for a new home or opportunities beyond our planet.

- **Significant of this project**

The "Spaceship System" project, created using OpenGL-GLUT, aims to create an immersive 3D experience by combining technical expertise with imaginative design. The project showcases the complexities of 3D modelling, animation, and interactive elements, demonstrating the capability of real-time rendering. The project focuses on human curiosity with space travel and the potential of futuristic planets, aiming to create narrative-driven graphics experiences that transcend still images. It pushes the limits of real-time rendering and animation, focusing on dynamic lighting effects and fluid animations for spaceship takeoffs, flights, and landings. The project also encourages user engagement through interactive components like spacecraft navigation and scene-specific features, offering a learning platform for understanding real-time 3D environments. The project highlights the collaboration between technical know-how and creative creativity, highlighting the limitless potential of computer graphics in art, technology, and imagination.

- **Target Application**

The "Spaceship System" project is a 3D simulation that combines storytelling with technology, offering an interactive experience in complex future settings. It can be used for research, education, entertainment, and technological demonstrations due to its adaptability.

- **Educational Applications**

The "Spaceship System" is a useful teaching tool for computer graphics, programming, and 3D design enthusiasts. It showcases advanced OpenGL-GLUT functionalities like texture mapping, dynamic lighting, and animation sequences, providing a hands-on learning experience. It can also be used in academic settings to demonstrate real-time rendering techniques and scene composition. The interactive nature encourages experimentation, fostering creativity and problem-solving skills in students studying these subjects.

- **Entertainment**

The project offers a visually immersive experience suitable for the entertainment industry, particularly in gaming and simulations. It features interactive spaceship navigation and diverse futuristic scenes, making it suitable for VR or AR platforms. The seamless transitions, dynamic lighting effects, and detailed textures keep users engaged. The project can be expanded with mission objectives, challenges, or puzzles to enhance its entertainment value.

- **Simulation and Training**

The "Spaceship System" can be utilized in simulations and training programs, particularly in aerospace, urban planning, and environmental studies. It can simulate practical tasks like docking scenarios and orbital mechanics, while also offering a platform for hypothetical scenarios.

- **Technical Demonstration Platform**

The project showcases OpenGL-GLUT's capabilities in rendering complex scenes and managing real-time interactions, showcasing advanced features like dynamic lighting and smooth animations. It's ideal for conferences, workshops, and technology exhibitions, allowing developers to benchmark hardware performance.

- **Art and Design**

The application's artistic appeal extends to digital art and design, featuring futuristic themes and creative elements like neon lighting and advanced civilizations. It inspires artists and designers to create their own 3D worlds, serving as a reference or starting point.

Knowledge gain from computer graphics

Computer graphics is a computer science field that involves creating images, animations, and simulations for various applications like gaming, movies, design, virtual reality, and education. It involves fundamental concepts and components.

Types of Computer Graphics

1. **Raster Graphics:**

- Composed of a grid of pixels (tiny dots of color) arranged in rows and columns.
- Commonly used in digital images, displays, and videos.
- Example: JPEG, PNG, BMP images.

2. **Vector Graphics:**
   - Based on mathematical formulas and geometric shapes like points, lines, and curves.
   - Scalable without loss of quality, making them suitable for logos and illustrations.
   - Example: SVG (Scalable Vector Graphics) files.

Graphics Pipeline

The graphics pipeline is a sequence of steps used to transform 3D objects into a 2D image on a screen. Key stages include:

1. **Modeling**:
   - Creating the geometric representation of objects using vertices and edges.
2. **Transformations**:
   - Applying translations, rotations, scaling, and other transformations to position objects in a 3D world.
   - Handled using matrices.
3. **Lighting**:
   - Calculating how light interacts with objects to produce realistic shading and shadows.
   - Types of lighting include ambient, diffuse, and specular lighting.
4. **Projection**:
   - Mapping the 3D scene onto a 2D plane (the screen).
   - Two types:
     - **Orthographic Projection**: No perspective, often used for technical drawings.
     - **Perspective Projection**: Mimics human vision by incorporating depth.
5. **Rendering**:
   - The process of converting 3D models and scenes into a final 2D image.
   - Techniques include ray tracing, rasterization, and real-time rendering.

Components of Computer Graphics

1. **Points, Lines, and Polygons:**
   - Basic building blocks for creating objects in a scene.
   - Polygons (often triangles) are widely used in 3D modeling.

2. **Textures:**

- Images applied to the surface of 3D objects to enhance detail and realism.
- Example: Applying a brick texture to a wall model.

3. **Color Models:**

- Used to represent colors in computer graphics.
- Common models:
  - **RGB (Red, Green, Blue)**: For screens.
  - **CMYK (Cyan, Magenta, Yellow, Key/Black)**: For printing.

Name of tools are used in this project

1. void update(int value)

- **Purpose**:
  - Updates the position of the moving object (a quad with a triangular attachment) in the scene.
  - Ensures continuous animation by repeatedly calling itself using glutTimerFunc.
- **Functionality**:
  - Checks if the position variable (representing the object's horizontal position) goes out of bounds (<-1.0).
    - If so, it resets position to 1.0f, restarting the object's movement from the other side of the screen.
  - Decreases the position value by speed to move the object leftward.
  - Calls glutPostRedisplay to redraw the updated position on the screen.
  - Sets a timer (glutTimerFunc(100, update, 0)) to call itself after 100 milliseconds for the next update.

2. void init()

- **Purpose**:
  - Initializes OpenGL settings, specifically the background color of the display window.
- **Functionality**:
  - Uses glClearColor(0.0f, 0.0f, 0.0f, 1.0f) to set the background color to black (RGBA: 0 for red, green, and blue, 1 for alpha/opacity).

3. void display()

- **Purpose**:
  - Renders the objects in the OpenGL window.

- **Functionality**:

  1. **Clearing the Screen**:

     • Clears the color buffer with glClear(GL_COLOR_BUFFER_BIT) to ensure the screen is refreshed before drawing.

     • Sets the current matrix to identity with glLoadIdentity() to reset transformations.

  2. **Drawing the Moving Object**:

     • Pushes the current transformation matrix using glPushMatrix().

     • Translates the object horizontally using glTranslatef(position, 0.0f, 0.0f). The position variable is updated dynamically to create the animation effect.

     •Draws two shapes:

       • **Quad (Rectangle)**:

         • A red-colored quadrilateral is drawn using glBegin(GL_QUADS) with the specified vertex coordinates.

       • **Triangle**:

         • A blue-colored triangle is drawn using glBegin(GL_TRIANGLES) to extend the object, resembling a simple rocket or arrow.

  3. **Restoring Transformation Matrix**:

     • The transformation matrix is restored using glPopMatrix().

  4. **Finalizing Rendering**:

     • Calls glFlush() to ensure all drawing commands are executed immediately.

4. void sound()

- **Purpose**:

  - Plays a sound file in the background to enhance the experience.

- **Functionality**:

  - Uses PlaySound() from the Windows API to play a sound file named a.wav in a loop.
  - The parameters passed to PlaySound():

    • "a.wav": Specifies the name of the sound file.

    • NULL: Specifies no additional flags.

    • SND_ASYNC | SND_FILENAME | SND_LOOP: Specifies that the sound is played asynchronously (does not block program execution), the sound is loaded from a filename, and it loops continuously.

5. int main(int argc, char argv)**

- **Purpose**:
  - The entry point of the program, setting up the OpenGL environment and starting the animation loop.
- **Functionality**:

  1. Initializes GLUT with glutInit(&argc, argv).

  2. Specifies the window size (320 x 320 pixels) and position (50, 50 from the top-left of the screen) using:

     - glutInitWindowSize(320, 320)
     - glutInitWindowPosition(50, 50)

  3. Creates a window titled "Translation Animation" with glutCreateWindow().

  4. Sets the display function using glutDisplayFunc(display), linking the display() function to render the scene.

  5. Initializes the OpenGL settings with init().

  6. Starts the timer for updates with glutTimerFunc(100, update, 0).

  7. Plays the background sound with sound().

  8. Enters the main event loop with glutMainLoop() to handle rendering and input continuously.

Knowledge applied in the filed

Computer graphics is a versatile field with diverse applications across various industries, offering opportunities in the job market and higher education. Applying computer graphics knowledge can lead to various career paths.

Job Market Purpose

1. **Industries That Demand Computer Graphics Skills**
   - **Entertainment & Media**: Game development, animation, VFX, and film production rely heavily on 3D modeling, rendering, and animation. Tools like Blender, Maya, and Unreal Engine are widely used.
   - **Software Development**: Many software products, especially in visualization, require computer graphics for interactive user interfaces, data visualization, and simulation tools.
   - **Architecture and Design**: Knowledge of CAD software and 3D rendering tools is essential for creating realistic architectural models and simulations.
   - **Healthcare**: Medical imaging and visualization systems use computer graphics for diagnostics, surgical simulations, and training.
   - **Virtual Reality (VR) and Augmented Reality (AR)**: Industries like gaming, real estate, and training are using VR/AR for immersive experiences.

2. **Job Roles**

    - Graphics Programmer/Software Developer
    - Game Designer/Developer
    - 3D Artist/Animator
    - UI/UX Designer
    - Visualization Specialist (e.g., in data science or engineering)
    - AR/VR Developer
    - Research Assistant in graphics-related technology firms

3. **Technical Skills in Demand**

    - Programming languages: C++, Python, GLSL, or JavaScript for WebGL
    - Graphics APIs: OpenGL, DirectX, Vulkan
    - Tools: Unity, Unreal Engine, Blender, Houdini, Adobe Creative Suite
    - Machine learning for graphics: Generative Adversarial Networks (GANs), neural rendering, etc.

4. **Freelancing and Entrepreneurship**

    - Freelancing platforms like Upwork and Fiverr offer opportunities in graphic design, 3D modeling, and animation.
    - Start your own venture in 3D printing, game development, or visual content creation.

Higher Study Purpose

1. **Specialized Areas in Computer Graphics**
    - **Rendering Techniques**: Study advanced methods like ray tracing, real-time rendering, or physically based rendering.
    - **3D Modeling and Simulation**: Explore geometric modeling, simulation of physical phenomena, or procedural content generation.
    - **Computer Vision and Image Processing**: Dive into augmented reality, digital twins, and AI-driven image synthesis.
    - **VR/AR Development**: Research immersive environments, interaction techniques, or head-mounted display technologies.
    - **Machine Learning in Graphics**: Applications in creating realistic models, textures, and animations using AI.
2. **Popular Programs and Universities**
    - Master's programs in **Computer Graphics and Visualization**, **Game Development**, **Human-Computer Interaction**, or **Media Technology**.
    - Institutions like Carnegie Mellon University, MIT, Stanford, ETH Zurich, and universities offering specialized graphics tracks.
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3. **Research Opportunities**

    - Engage in research projects focusing on graphics-related problems, such as real-time rendering, simulation, or AI for graphics.
    - Publish papers in leading journals or conferences like SIGGRAPH, Eurographics, or IEEE Visualization.

4. **Networking and Collaboration**

    - Attend international conferences and workshops like SIGGRAPH, GDC, or specialized VR/AR meetups.
    - Collaborate with professionals and academics to gain hands-on experience in state-of-the-art projects.

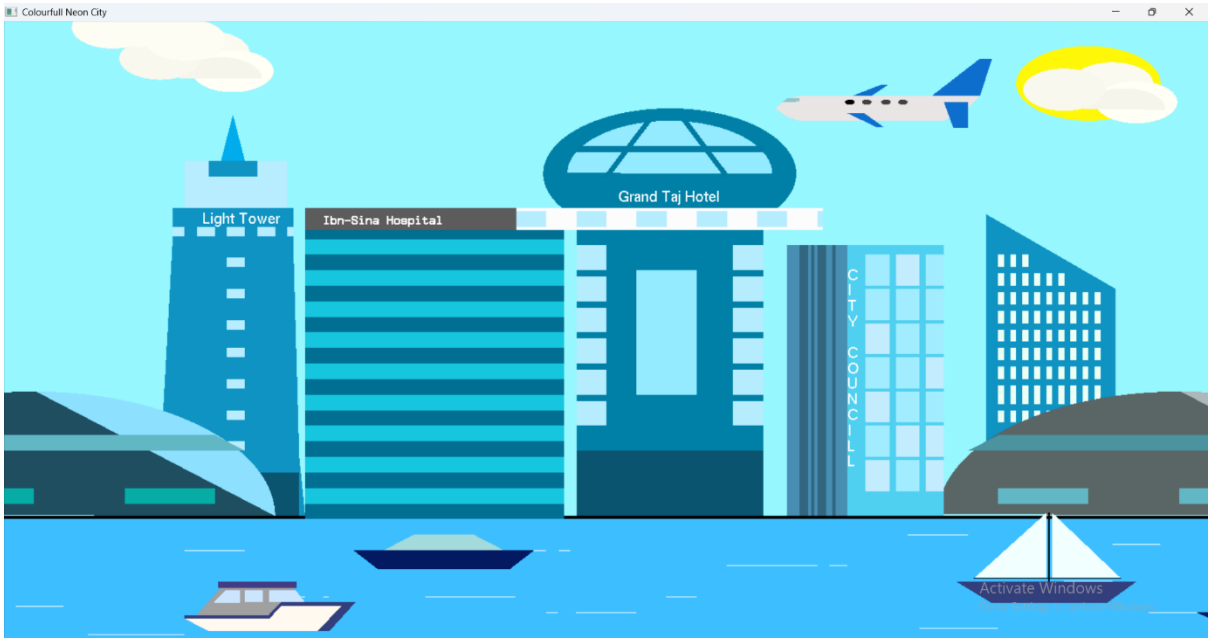
5. **Scholarships and Fellowships**

    - Many tech giants like NVIDIA, Adobe, and Google offer scholarships and fellowships for students researching computer graphics.

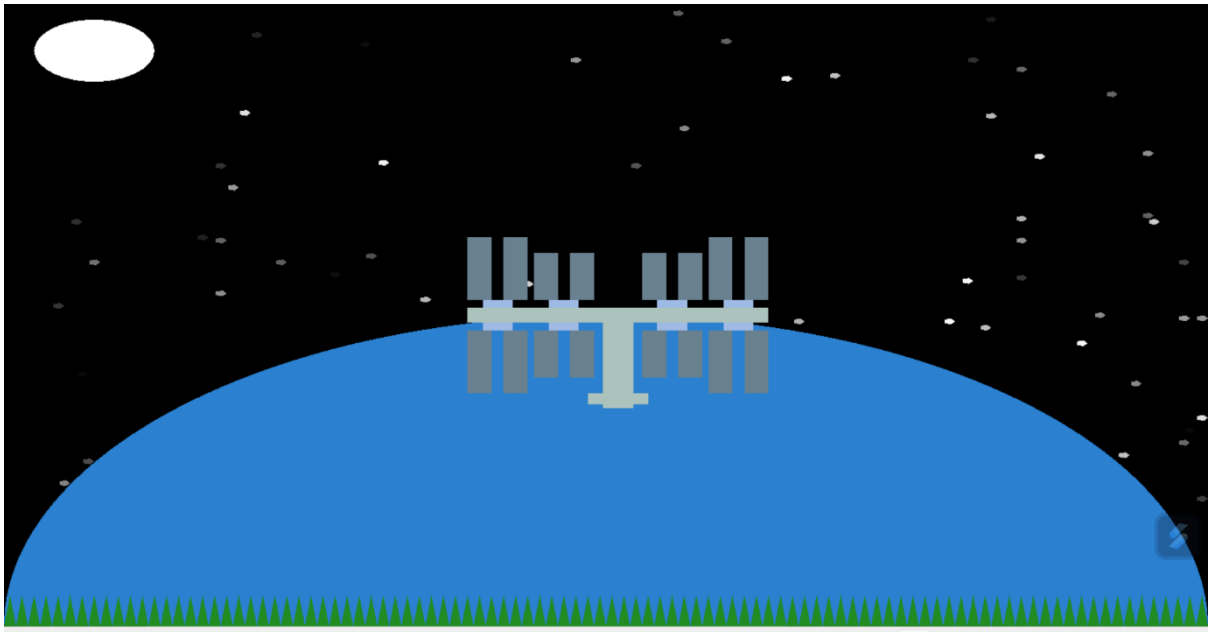
References

❖ YouTube Link: https://www.youtube.com/@abm_prottoy
❖ GitHub Link: https://github.com/abmprottoy

Screenshots of project



Picture 1



Picture 2

Picture 3